December 2001

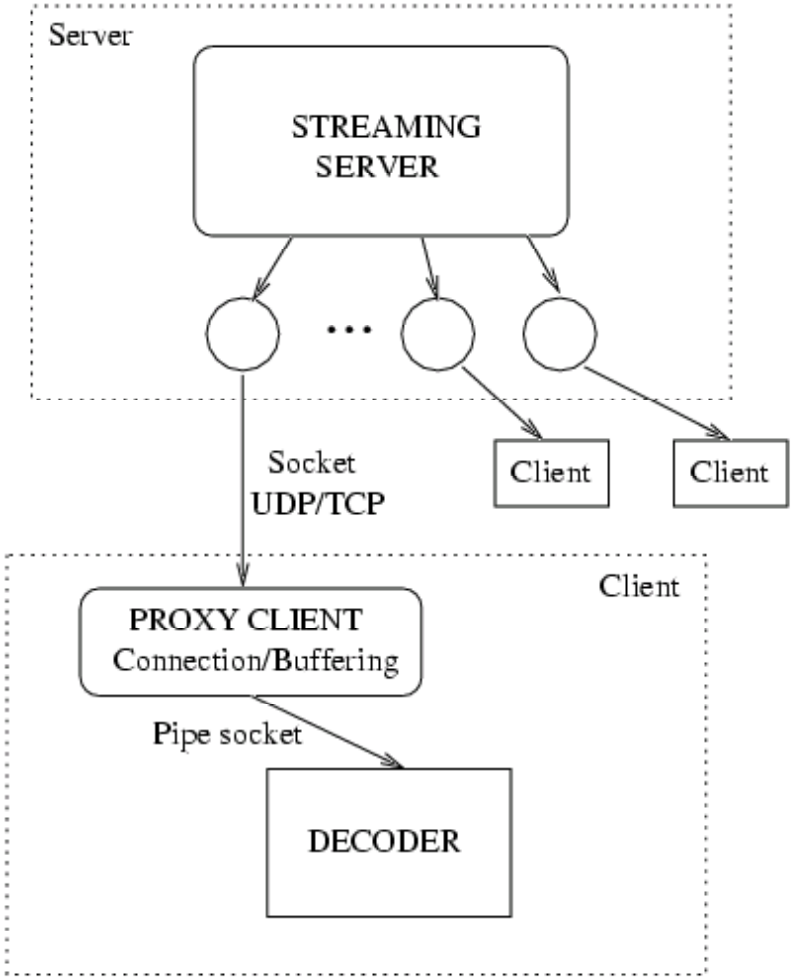
Multimedia Streaming on Wireless Networks

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System Architecture



Experimental Setups

- Streaming over TCP and UDP

- Eth -> Eth
- Eth -> Wireless
- Wireless -> Wireless
- Wireless -> Wireless Ad-Hoc

Experimental Results

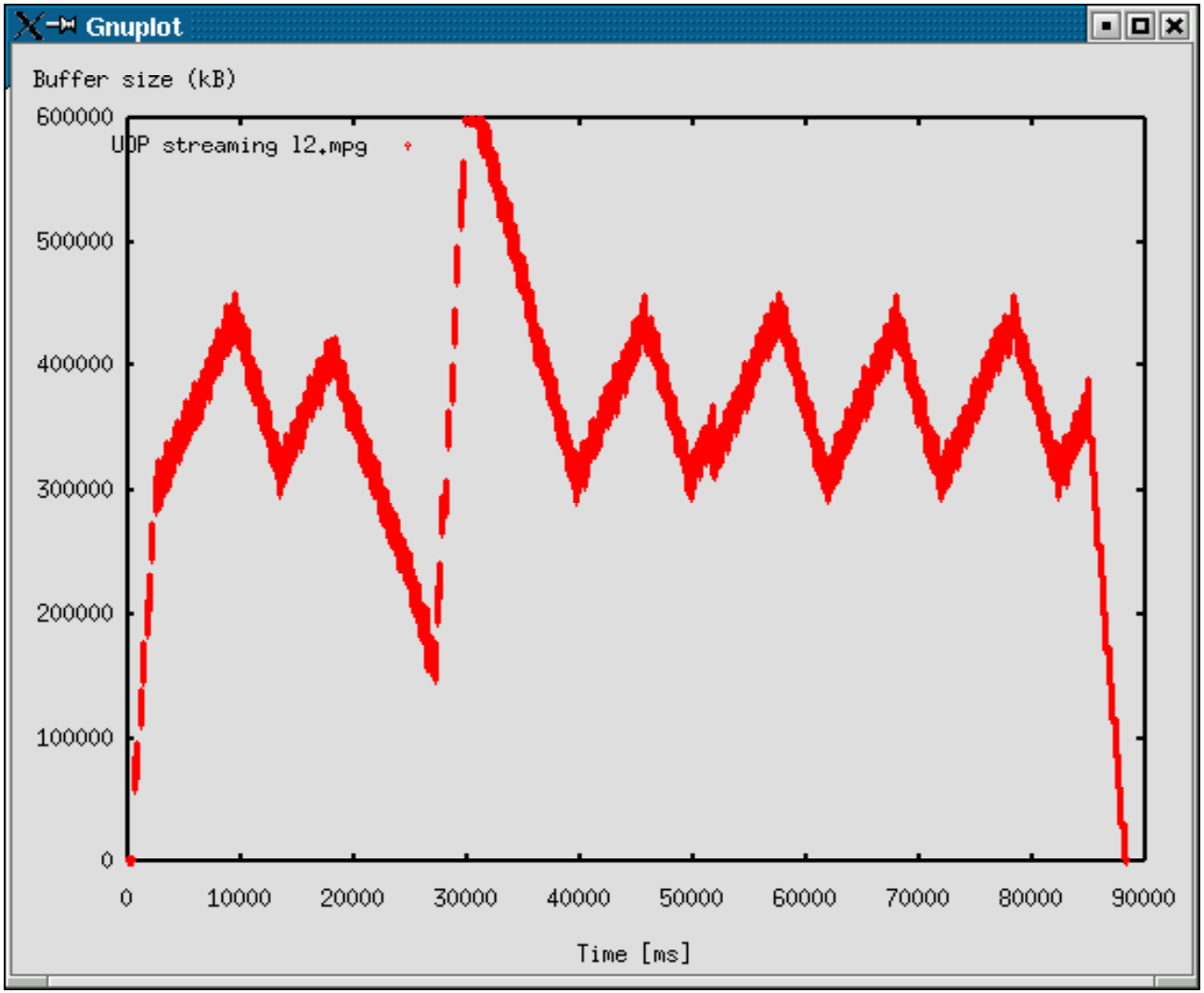
- Correctness validation:
  - TCP: pace and comparison tests
  - UDP: pace and continuity tests

- Performance characteristics
  - Initial latency
  - Drop rates
  - Network throughput
  - Hand-off latency

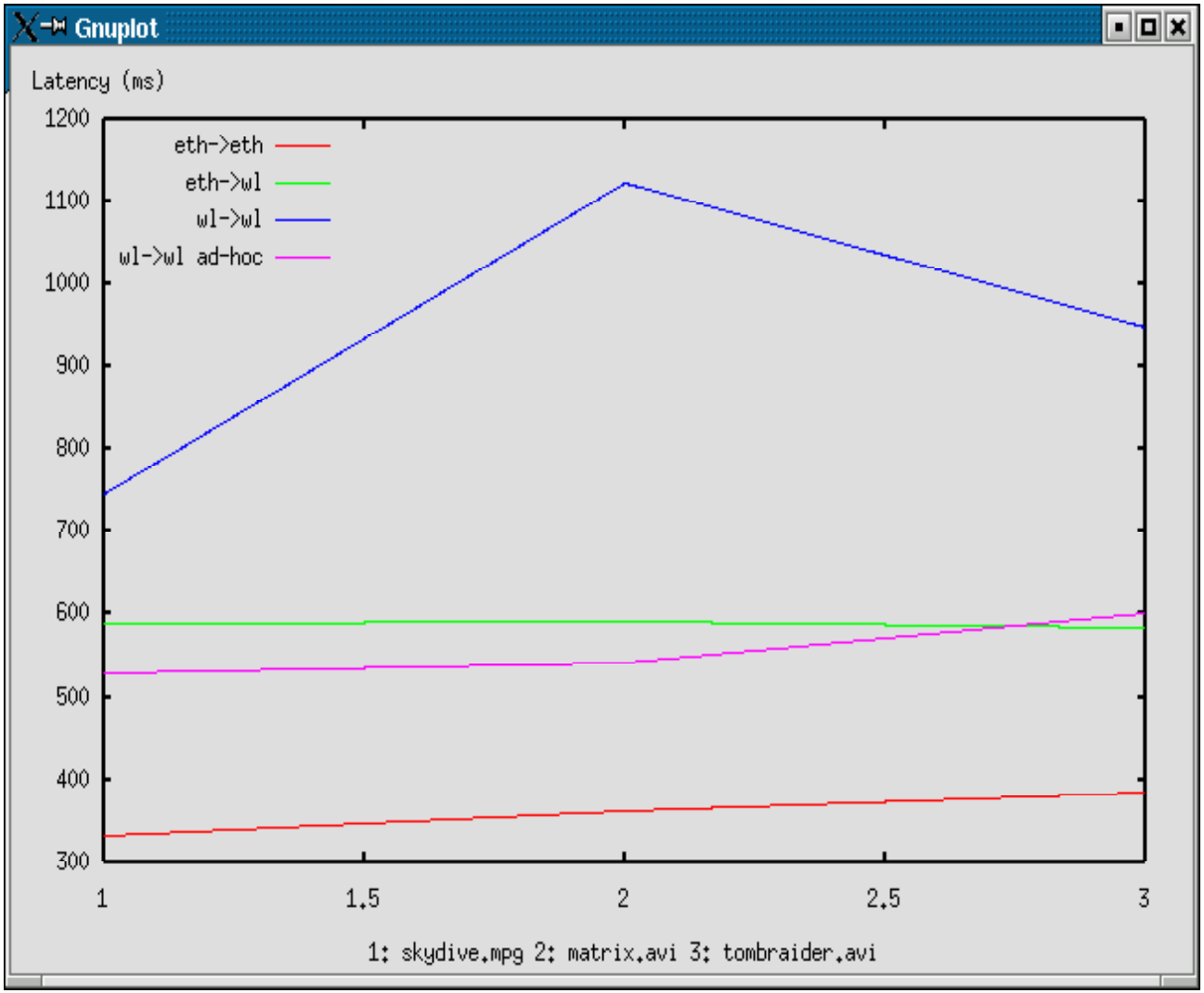
Pace Test

	et-et	et-wl	wl-wl	ad-hoc
tcp:	**[s]**	**[s]**	**[s]**	**[s]**
skydive.mpg	124.7	124.7	124.7	124.8
matrix.avi	150.1	149.6	149.6	149.6
tombraider.avi	118.1	118.1	118.1	118.1
udp:				
skydive.mpg	124.6	124.6	124.6	124.4
car5.mpg	25.8	25.8	25.8	25.8

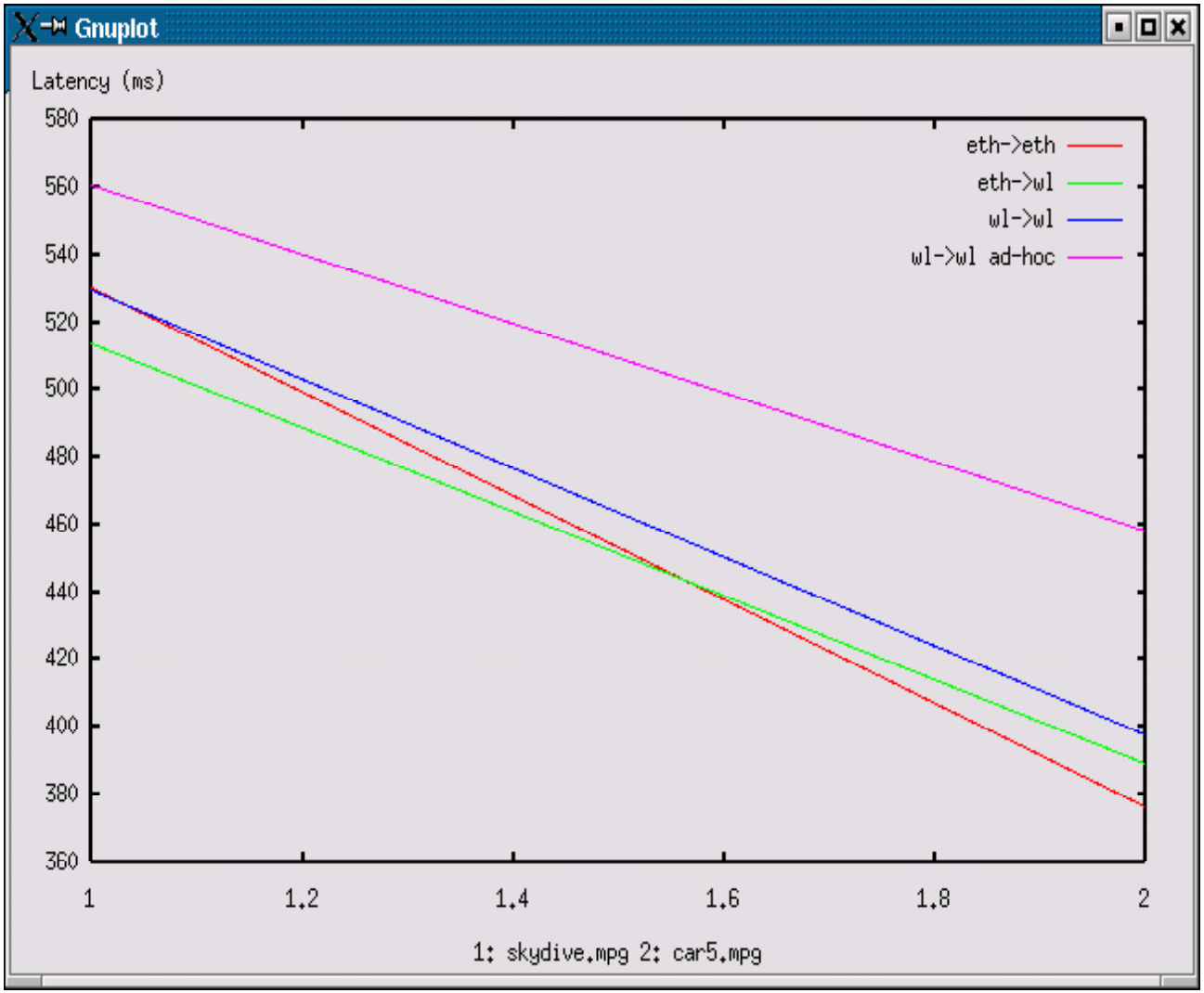
UDP Flow Control



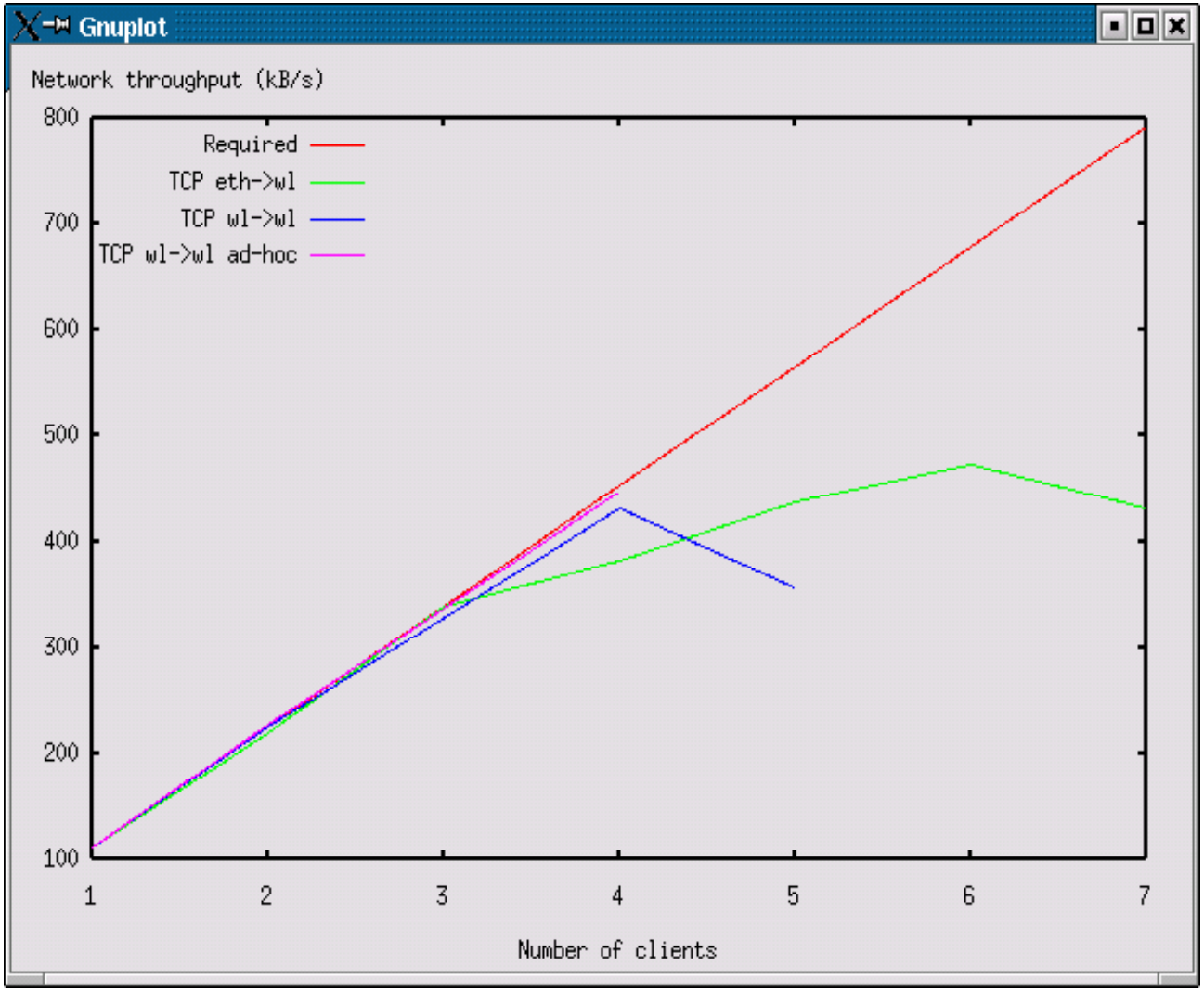
Initial Latency: TCP



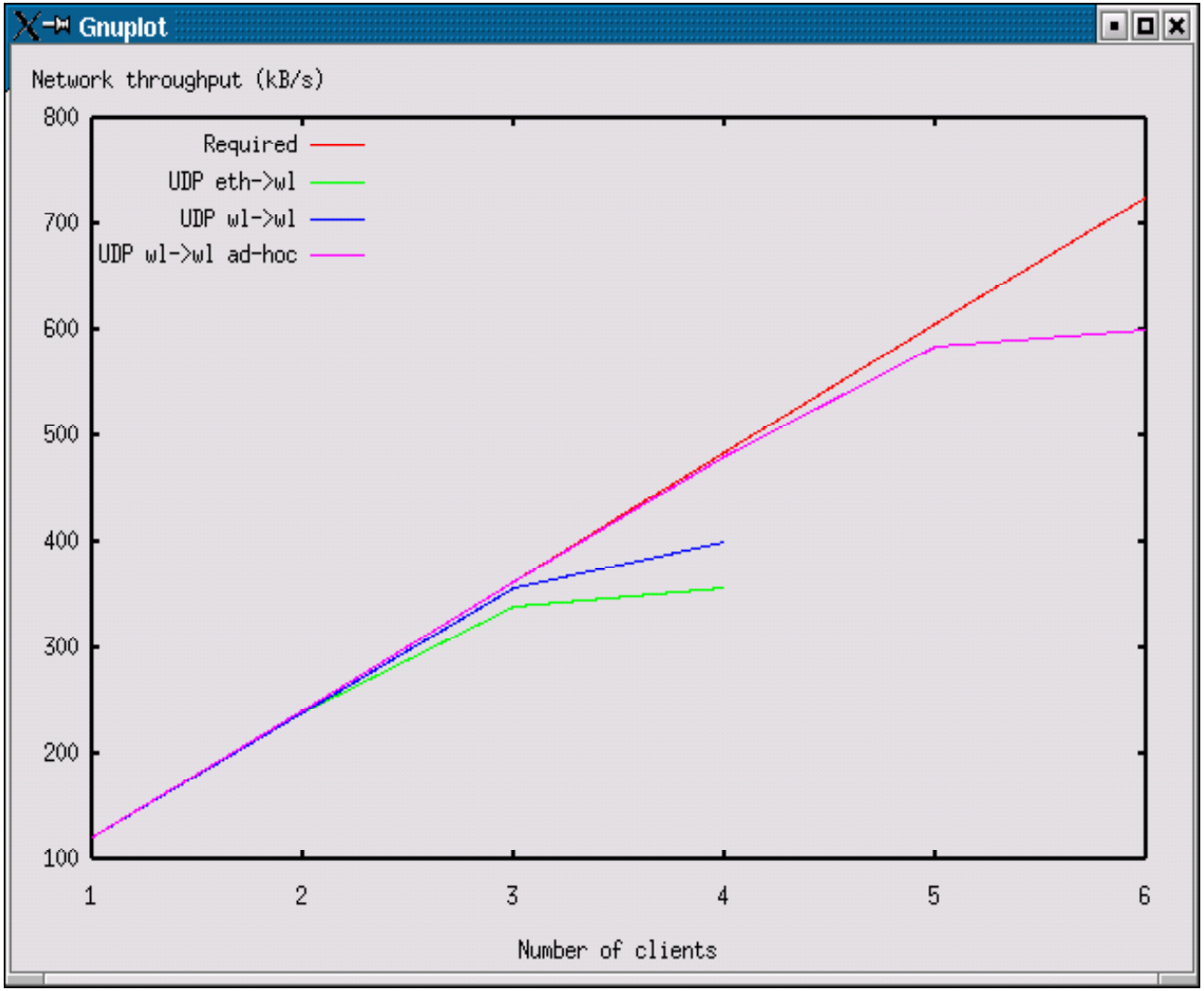
Initial Latency: UDP



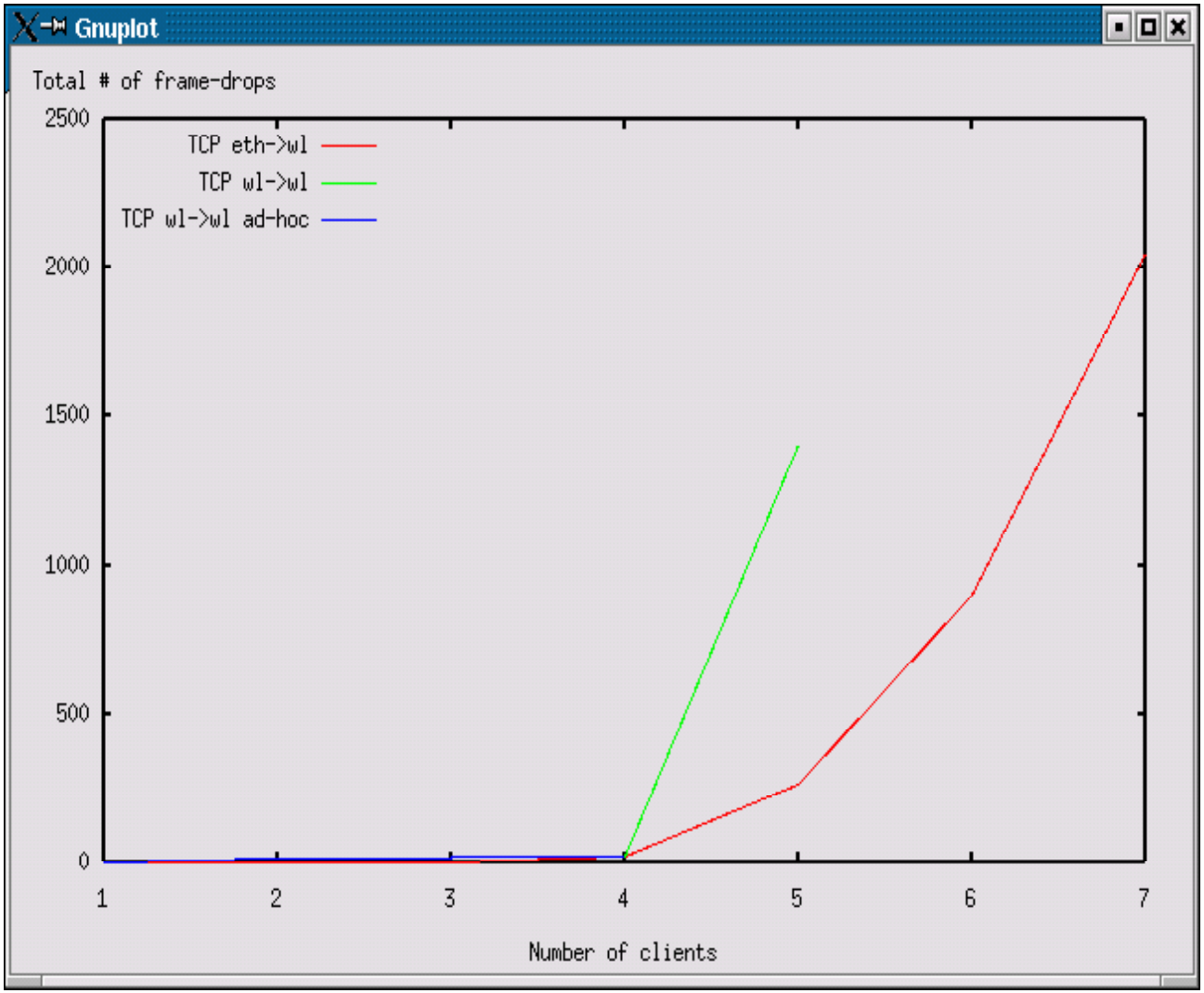
Network Throughput: TCP



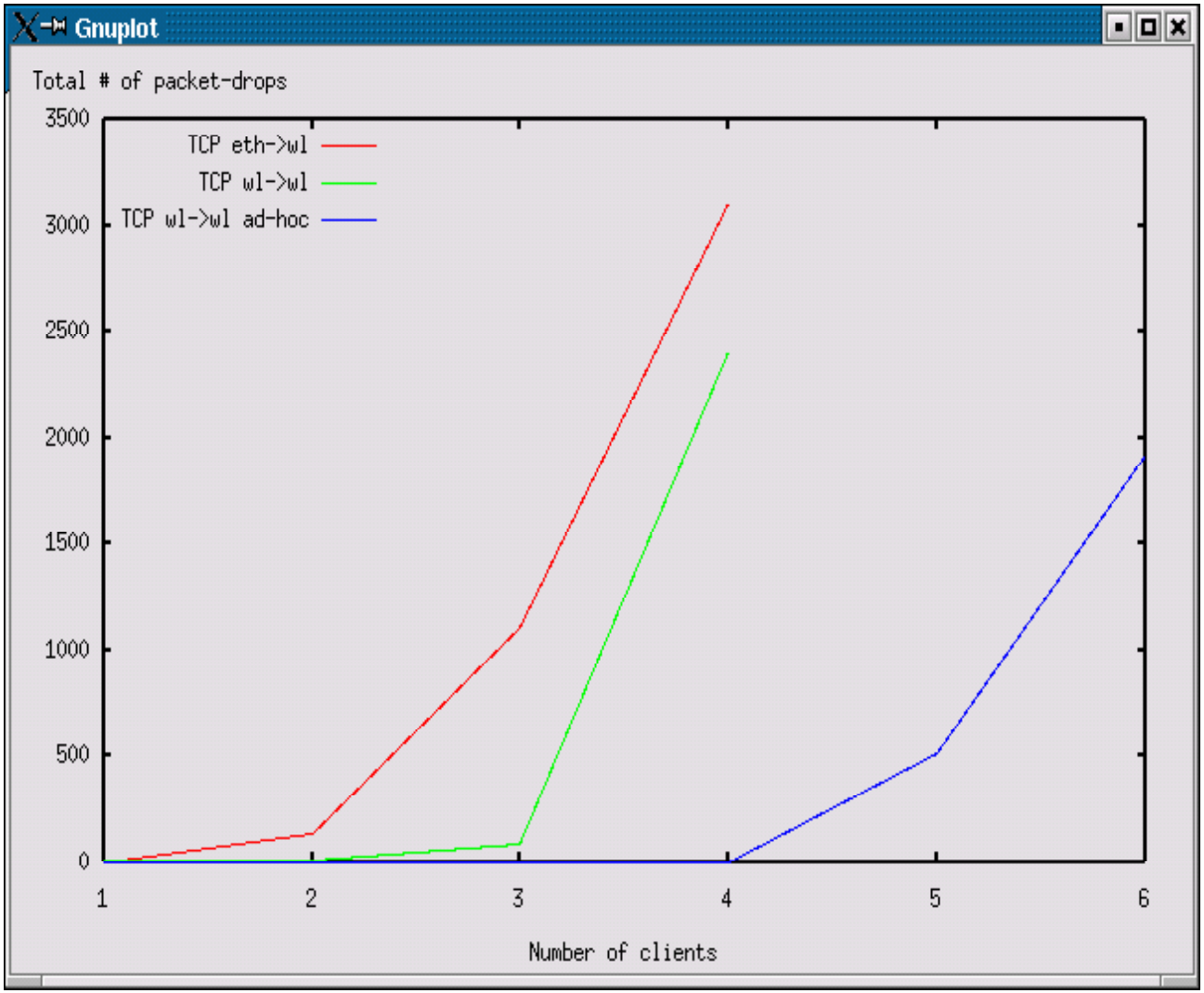
Network Throughput: UDP



Frame Drops: TCP



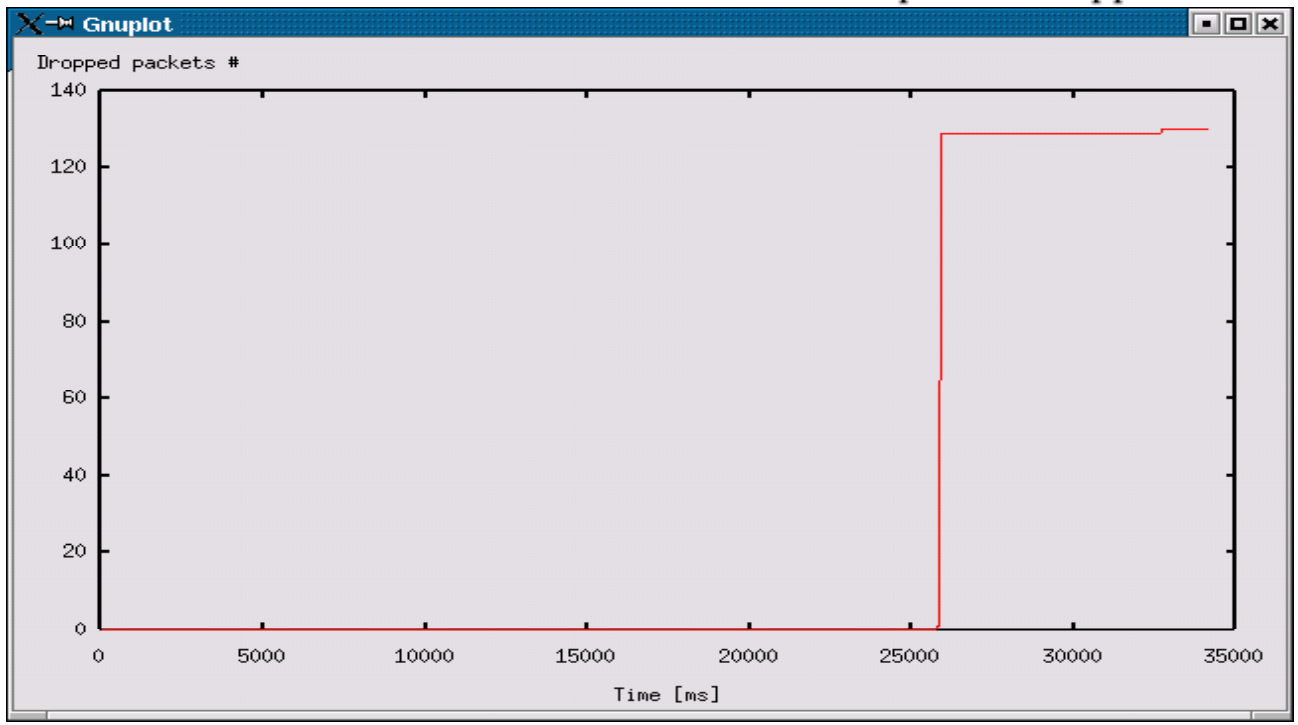
Packet Drops: UDP



Base-Station Hand-offs

Access points	Duration	Packets dropped
$00:02:2D:0D:A9:9F \rightarrow 00:02:2D:28:D6:2E$	$56ms$	53
$00:02:2D:28:D6:2E \rightarrow 00:02:2D:0D:A9:F5$	$54ms$	129
$00:02:2D:0D:A9:F5 \rightarrow 00:02:2D:0D:A9:9F$	$74ms$	15

Table 1: Hand-off duration and the number of packets dropped.



Conclusions

- Difference between setups:
  - Where are the packets dropped?
- UDP streaming:
  - Small initial latency
  - Throughput doesn't degrade under high load
- TCP streaming:
  - Does not inherently support real-time streaming
  - High socket set-up time
  - Throughput suffers under high load
- Packet dropped during hand-offs